

Quantum Rings

Final Presentation

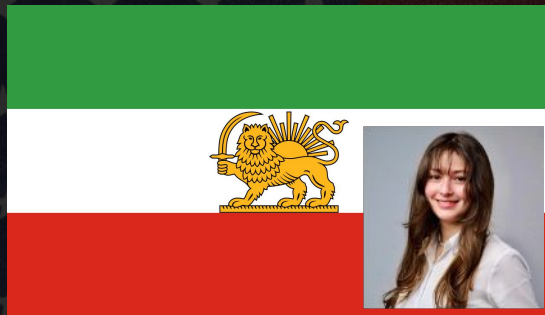
Jose Abuan, Melody Goldanloo, Paul
Lintzen, Aradhana Ramesh

Our Team



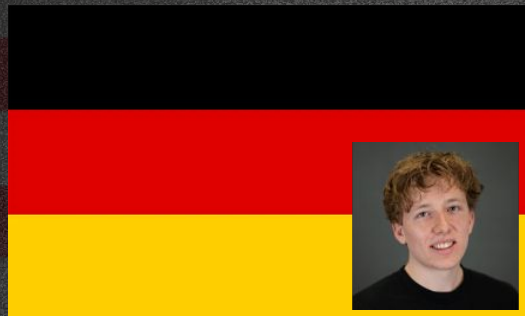
Jose Abuan

Rising Senior



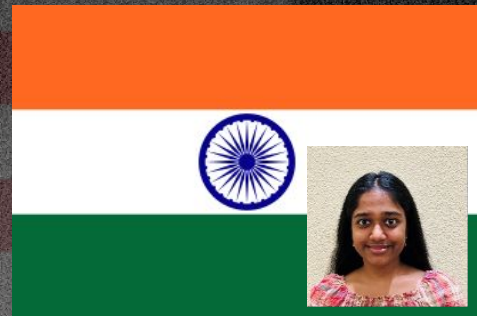
Melody Goldanloo

Rising Graduate Student



Paul Lintzen

Rising Graduate Student



Aradhana Ramesh

Rising Junior

$|000\rangle$ Intro to Quantum

$|001\rangle$ Quantum Rings

$|010\rangle$ Final Design

$|011\rangle$ Demo

$|100\rangle$ Summary

$|000\rangle$

Intro to Quantum

What Makes Quantum Computing Different?

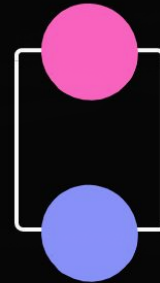
Classical computers use bits. Quantum computers use **qubits**.

Classical Bit	Qubit
0 or 1	0, 1, or both
Logic gates	Quantum gates
Stable and mature	Powerful, but noisy and fragile

BITS



QUBITS



SUPERPOSITION

Potential Uses

Optimization & Decision Systems

- Financial modeling
- Traffic and grid optimization
- Supply chain and inventory optimization

Science Simulations

- Drug discovery and development
- Weather forecasting and climate modeling
- Space system modeling

AI, Data, & Security

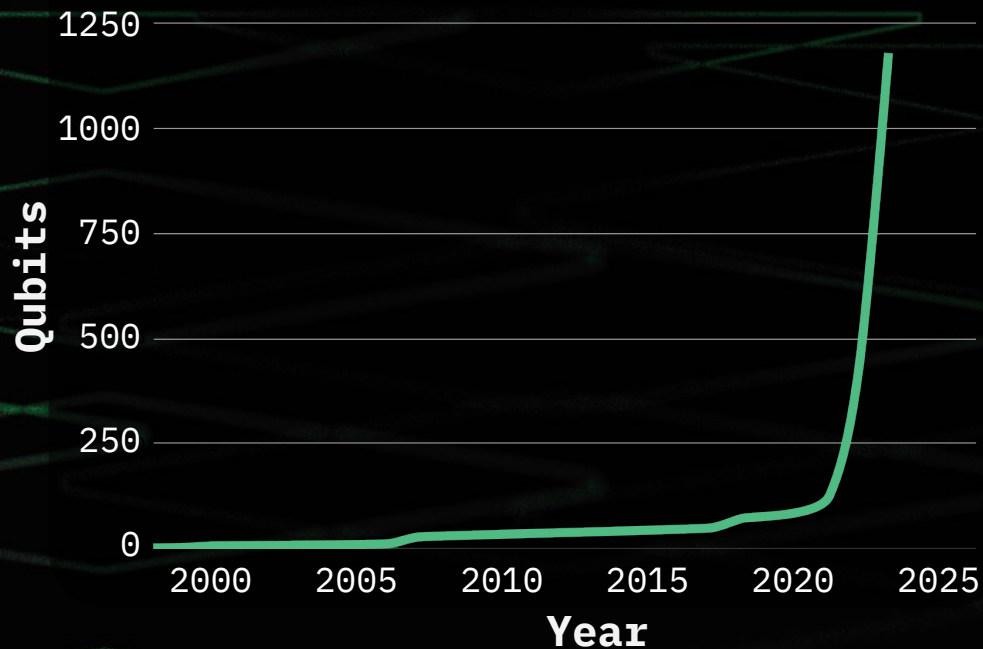
- AI and ML
- Cryptography and data security
- Disease risk predictions
- Anomaly/Threat detection

Quantum Computing is Still Young

The first *working* quantum computer was built in **1998**.

It used 2 qubits.

Gate-Based Quantum Computers:
Qubit Growth Over Time

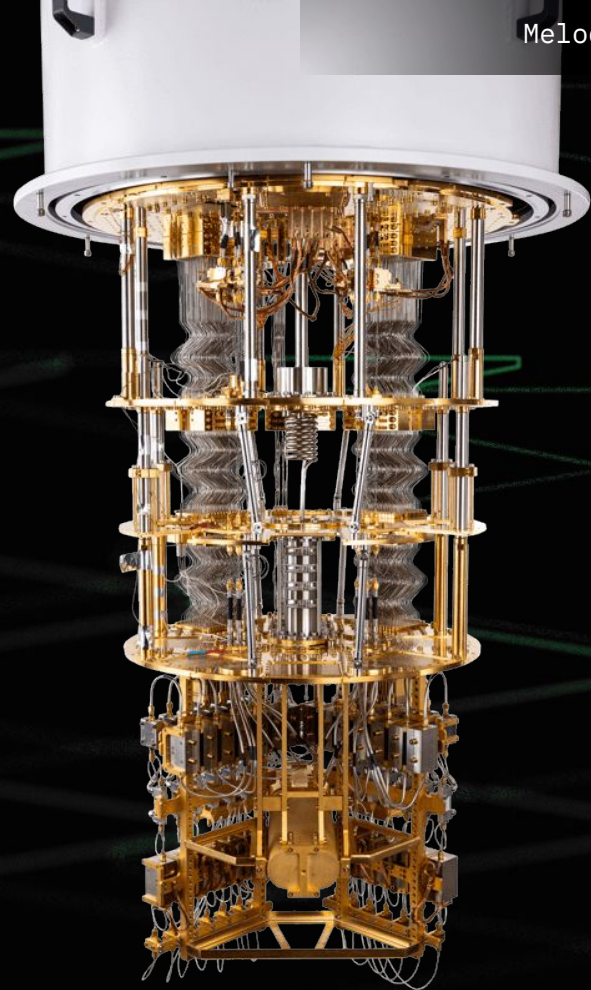


40+

commercially available QPUs worldwide.

Quantum Computers are Expensive

Dilution refrigerator	\$200k – \$1m+
QPU + Packaging	\$500k – \$500m+
Control electronics	\$500k – \$2m+
Facility + Installation	\$100k – \$500k
Upkeep + Staff	\$100k – \$1m/yr
Total	≥ \$1.5m



$|001\rangle$

Quantum Rings

Who Is Quantum Rings?

- Quantum startup from Boulder, CO
- Offer open software development kit
- Allows users to run quantum circuits on classical hardware instantly
- Expanding to offer cloud simulation

Why Simulate?

Quantum Hardware

Simulation



3/4 Drops

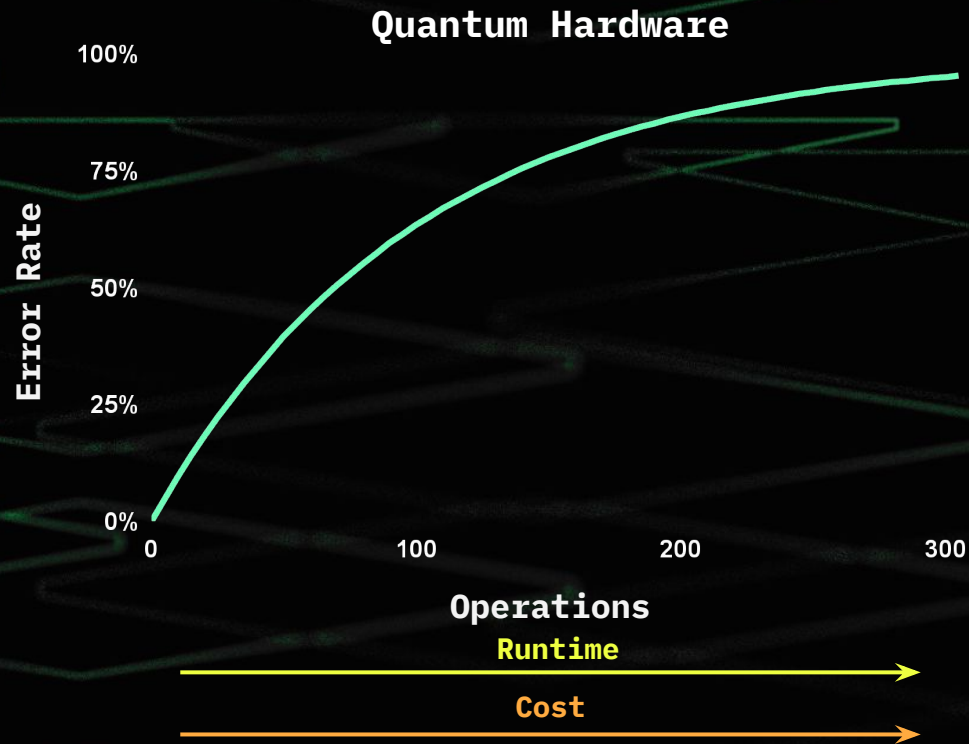


0/4 Drops

Quantum Hardware Today

Quantum Today:

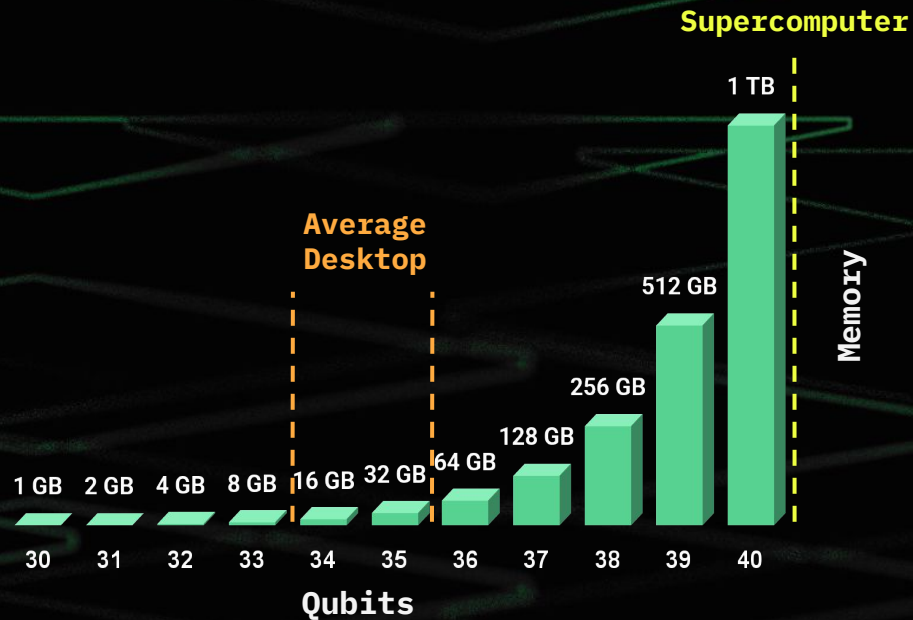
- Error rate grows with operations
- **\$100+** per minute for execution on QPUs
- Typical wait times on IBM **~12** hours



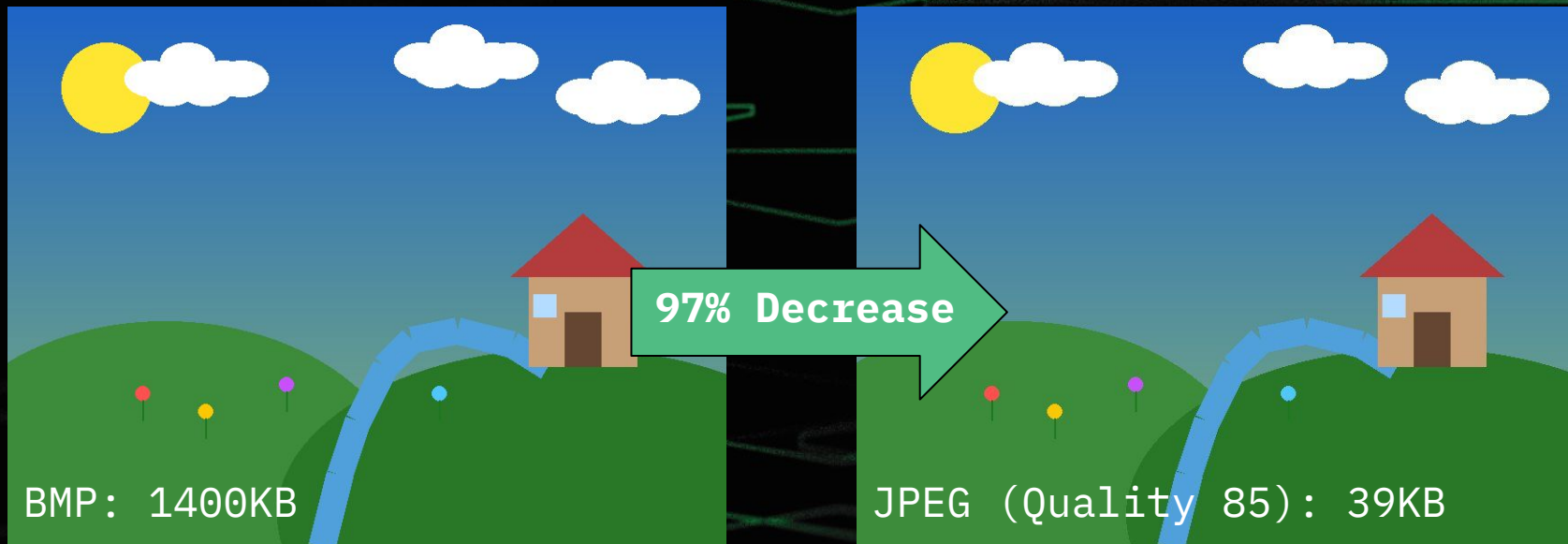
Simulation Problem

Traditional Simulation:

- Extremely memory intensive (2^n states for n qubits)
- Large circuits require cutting edge supercomputers



Compression



- JPEG Quality: ↓ smaller file, more artifacts ↑ larger file, better quality
- QR Threshold: ↓ lower mem/runtime, lower accuracy ↑ higher mem/runtime, better accuracy

Our Project



Predicting Simulation Time

Runtime/memory use depend heavily on the specific circuit.

Complex Circuit Structure:

- Cut pressure
- Parameter entropy
- Graph bandwidth
- Entanglement pattern

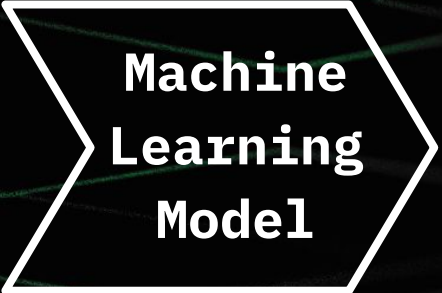
"A 30-qubit GHZ state finishes in seconds, while a 30-qubit deeply-entangled random circuit could run for hours and consume significant GPU memory"

- Quantum Rings

The Solution

Input:

- QASM circuit file
- Backend config & run settings



**Machine
Learning
Model**

Output:

- Runtime estimate
- Peak memory estimate

Quantum Assembly Language (QASM) Files

- Text format describing quantum gate operations
- Compiled and run on quantum hardware

Quantum Assembly

```
OPENQASM 2.0;
include "qelib1.inc";
qreg q[3];
creg c[3];
// Initialize superposition
h q[0];
// Entangle qubits
cx q[0], q[1];
cx q[1], q[2];
// Rotate qubit
ry(pi/4) q[2];
// Measure all
measure q -> c;
```

Classical Assembly

```
; Load values into registers
mov eax, 5
mov ebx, 3
; Arithmetic
add eax, ebx           ; eax = 8
imul eax, ebx          ; eax = 24
; Bitwise ops
and eax, 0xFF
shl eax, 2
; System call
mov eax, 1
xor ebx, ebx
int 0x80
```

Backend Configuration & Run Settings

Backend:

- Scarlet (CPU)
- Amber (GPU)
- Serin (Hybrid)

Precision:

- Single (32 bits)
- Double (64 bits)

Threshold:

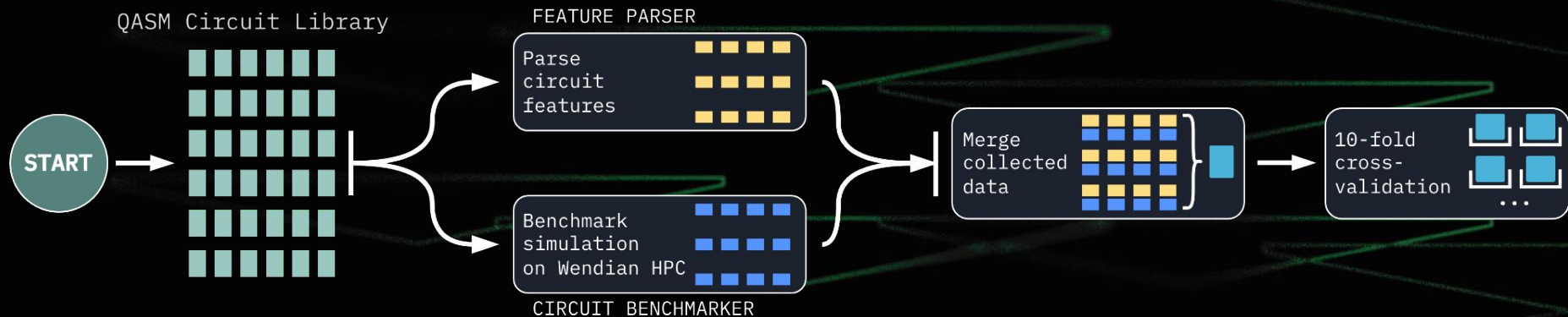
- 2 - 1024
(Efficiency vs
Accuracy Tradeoff)

$|010\rangle$

Final Design

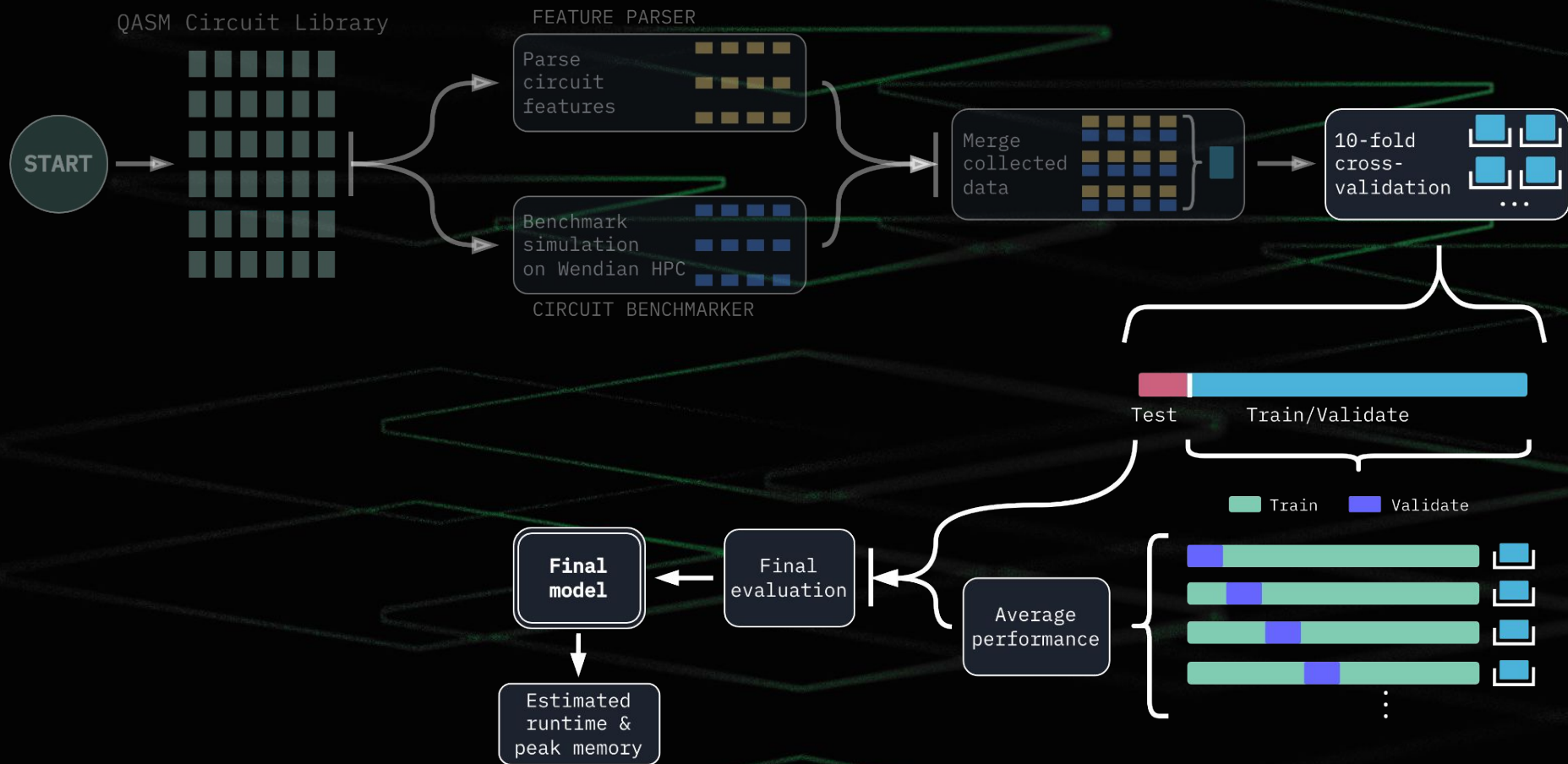
MODEL DESIGN

Melody



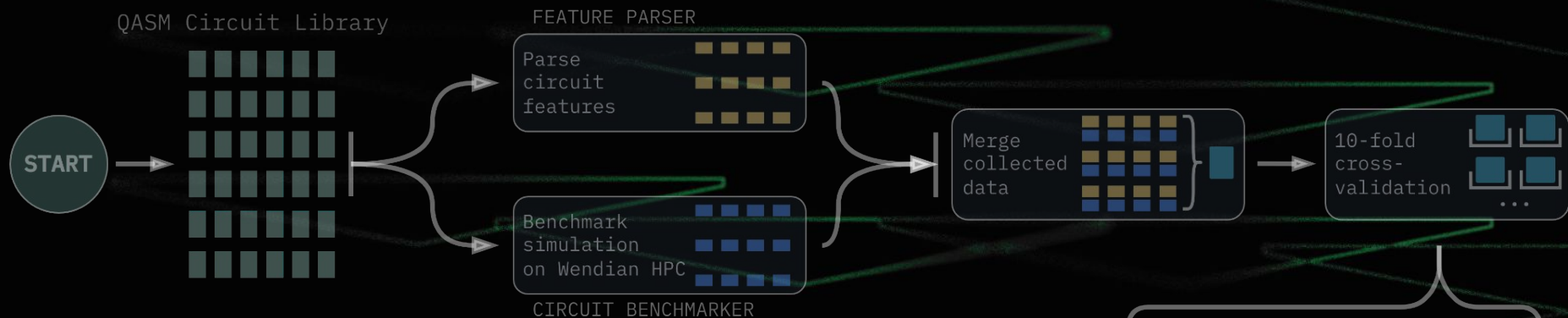
MODEL DESIGN

Melody

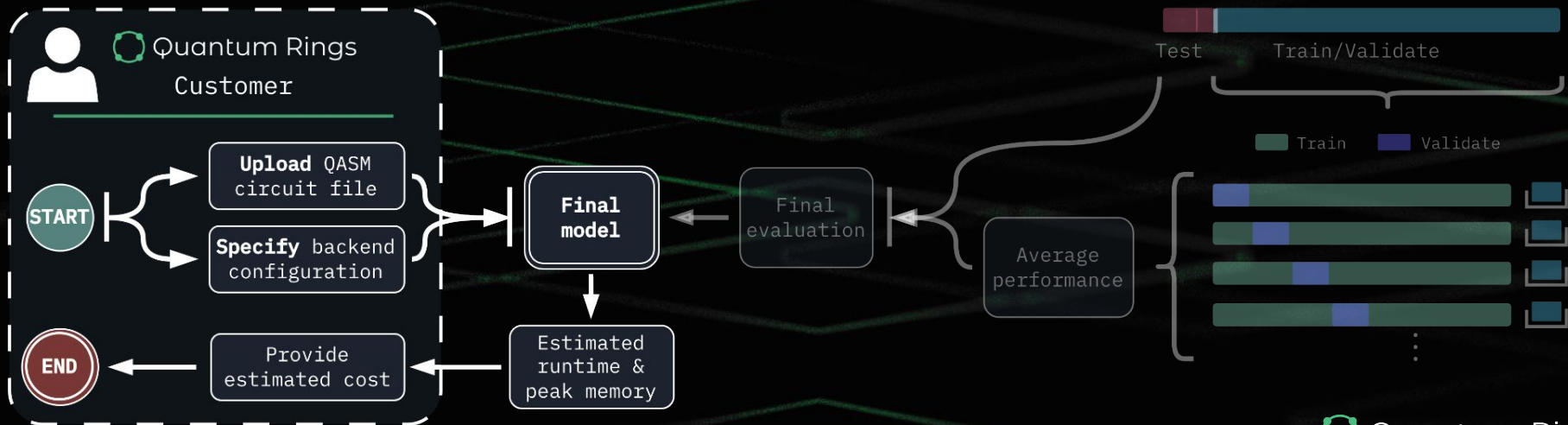


MODEL DESIGN

Melody



FINAL PIPELINE

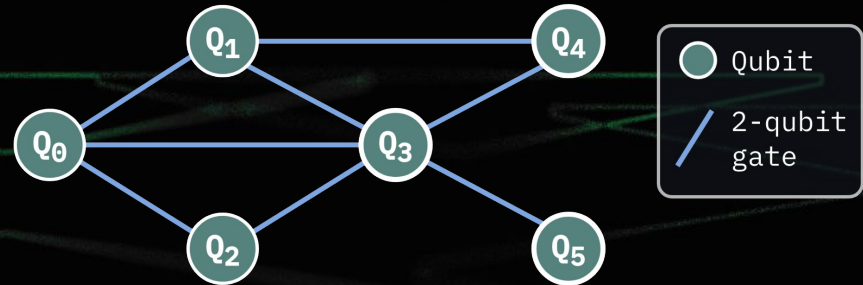


Feature Parser Architecture

Simple Features

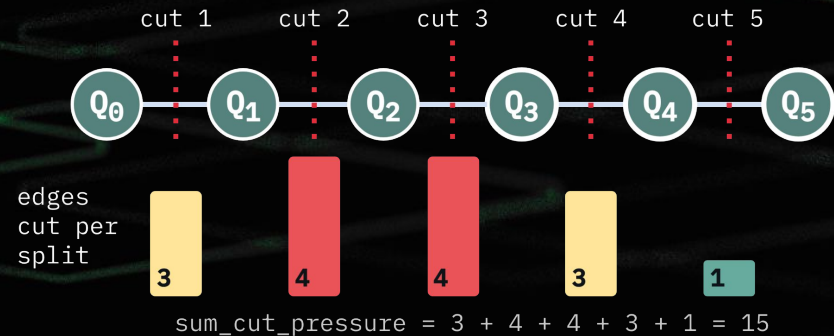
Examples: n_qubits , n_cbits ,
 n_tdg , ...

Interaction Graph

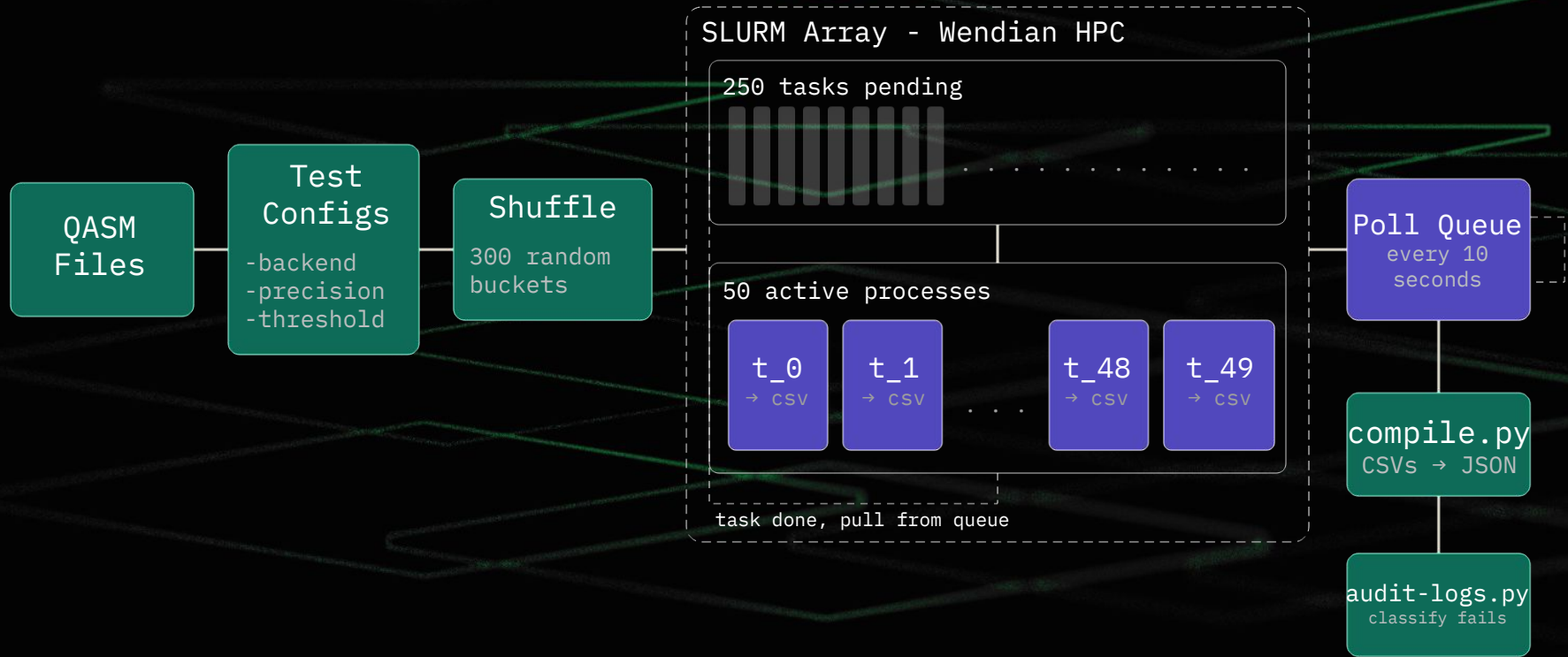


Complex Features

$n_edge_repetitions$,
 $crude_depth$, $cut_pressure$, ...



Benchmarking Architecture



Final Dataset: 790 Unique Circuits

Fault-Tolerant Algorithms

Examples: cryptography, search algorithms

Quantum Error Correction (QEC)

Examples: syndrome checks, code circuits

Noisy Intermediate-Scale (NISQ)

Examples: optimization, variational algorithms

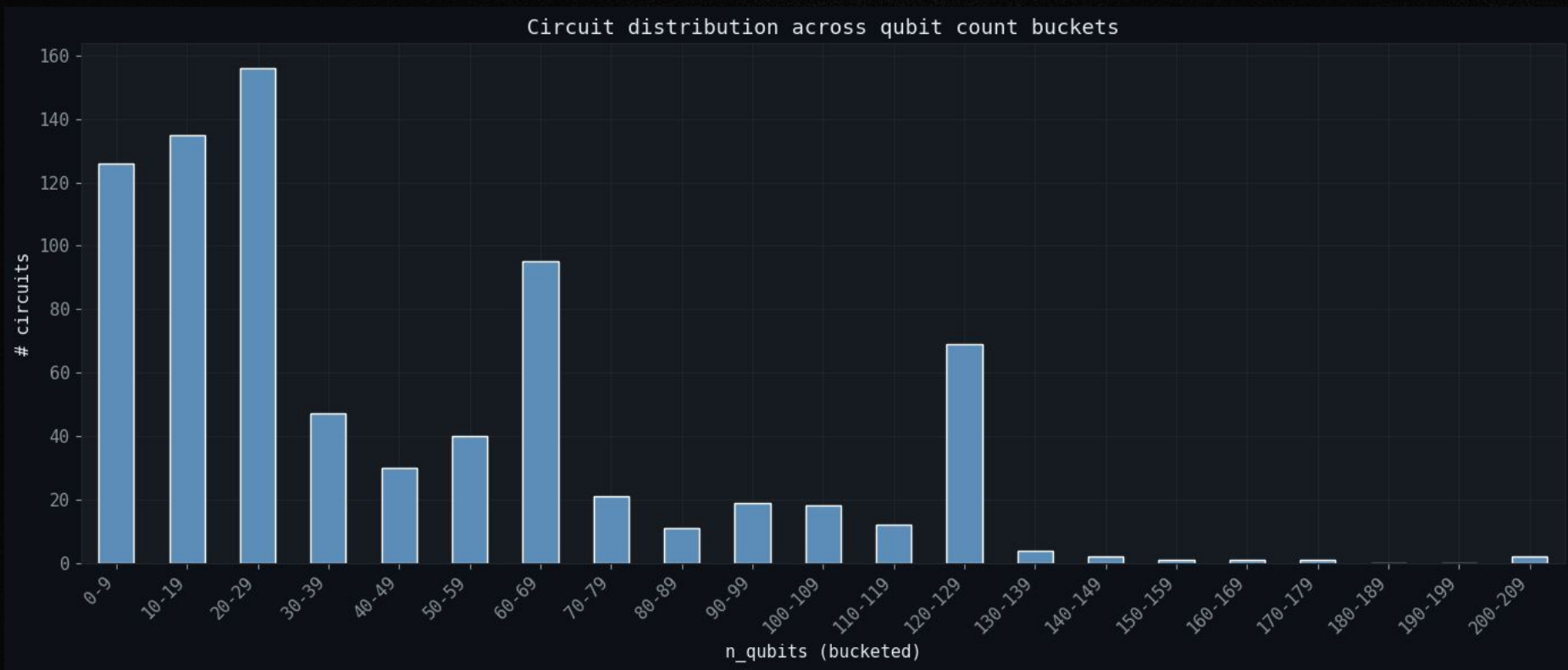
Synthetic Benchmarks

Examples: random circuits, generated test cases

A

Features

Qubit Counts Across 790 Circuits



140 Total Features

Extracted + Engineered

30 Gate Counts

- n_cx
- n_h

108 Circuit Specific

- n_edge_repetitions
- sum_cut_pressure

1 Numeric

- Threshold

2 Categorical

- Precision
- Backend

Key Features

Feature Importance by SHAP (Mean |SHAP value|)

Runtime (s)



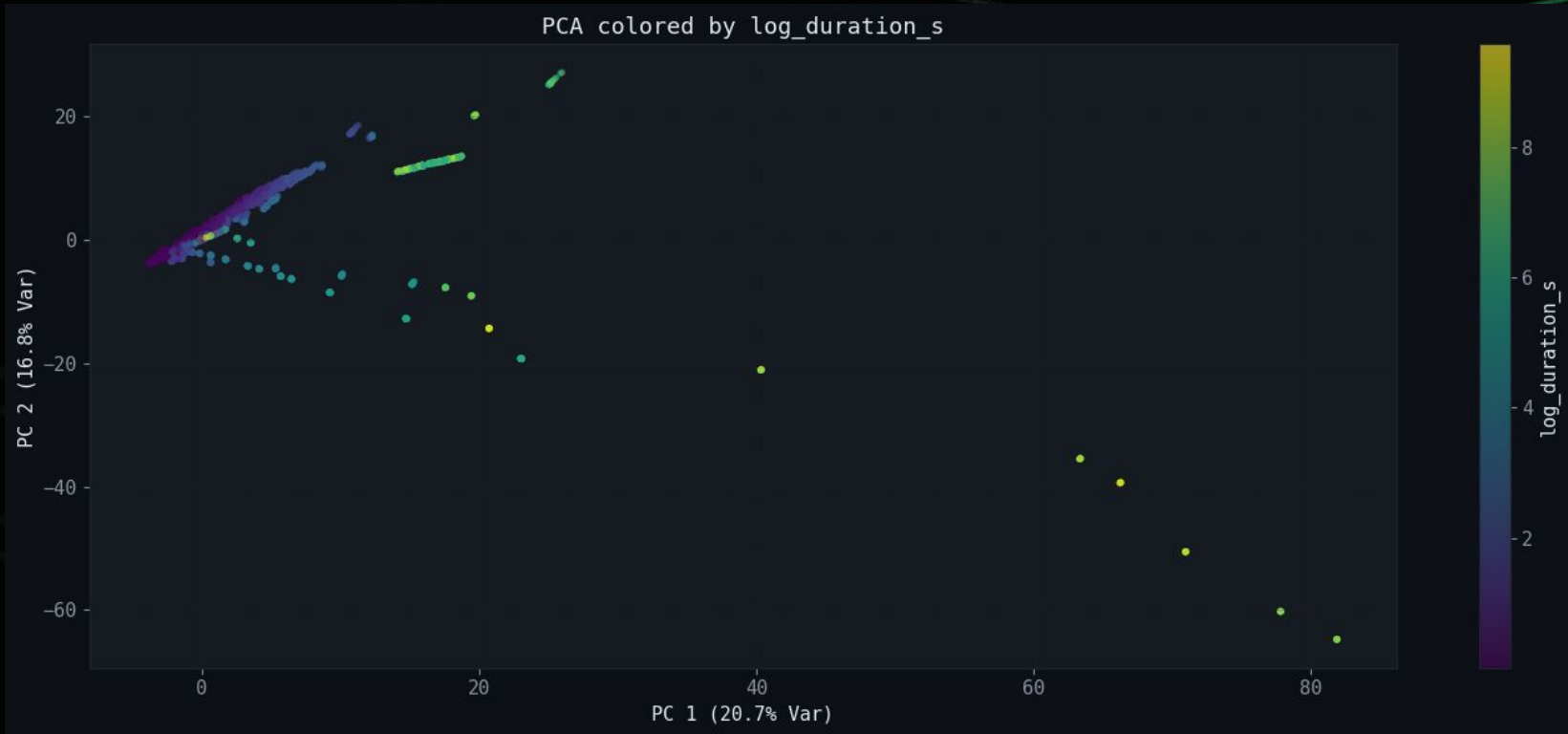
Peak Program Memory (MB)



Peak Simulation Memory



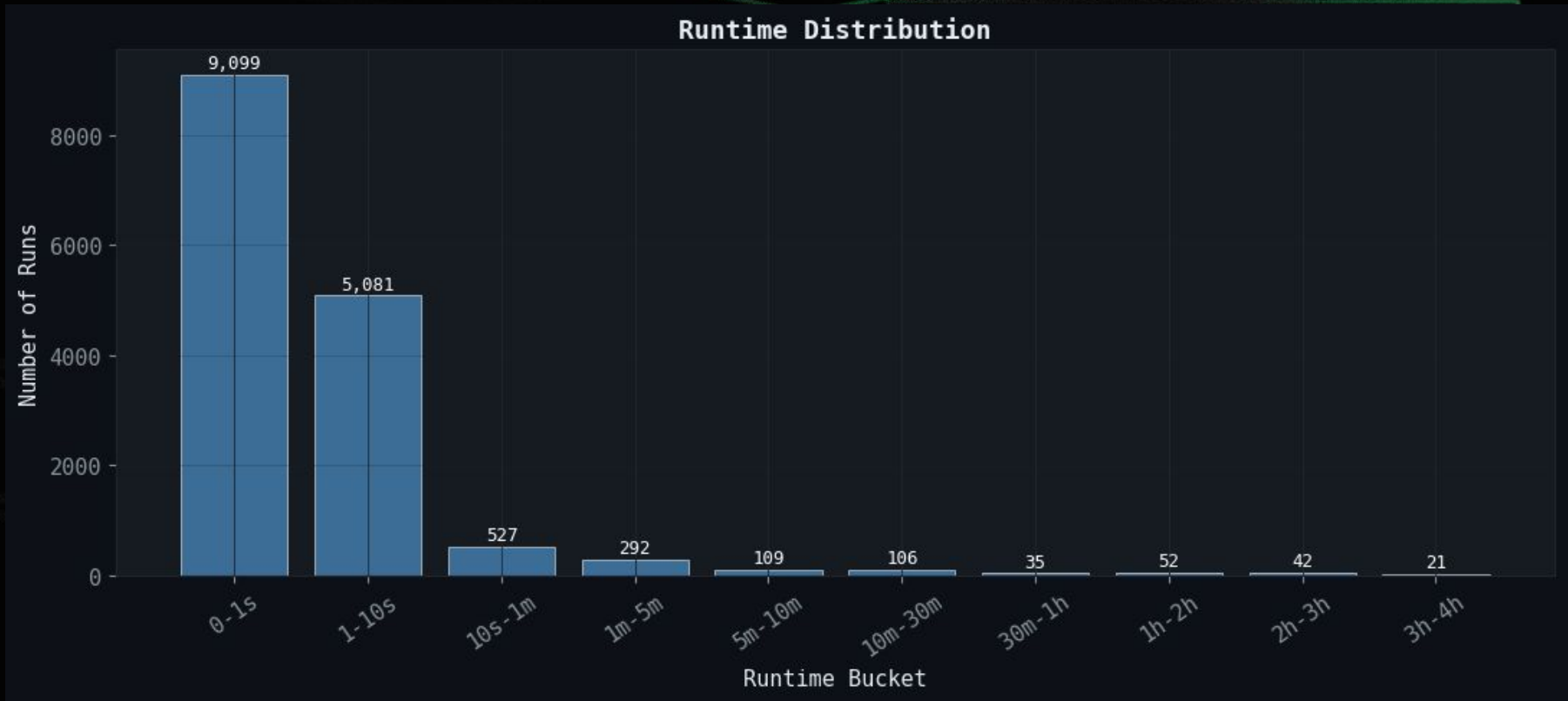
Principal Component Analysis



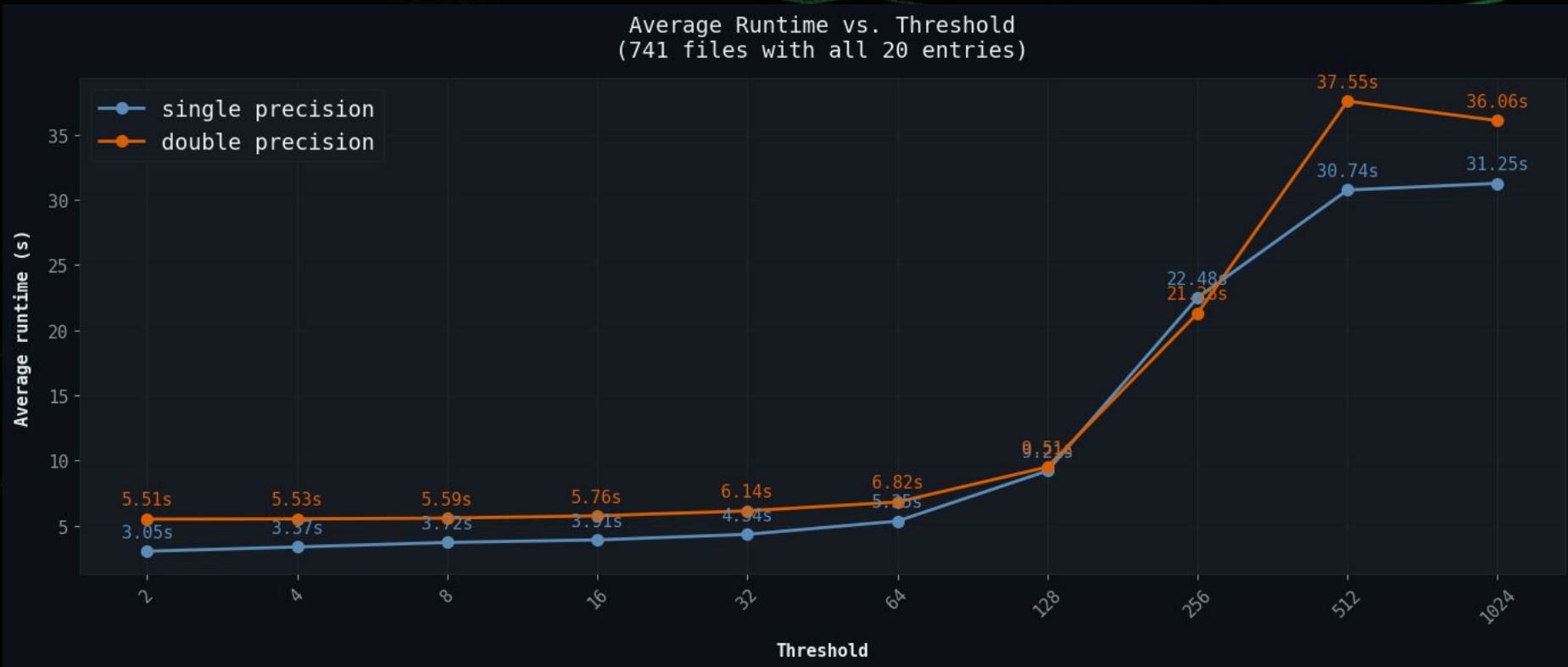
B

Runtime Results

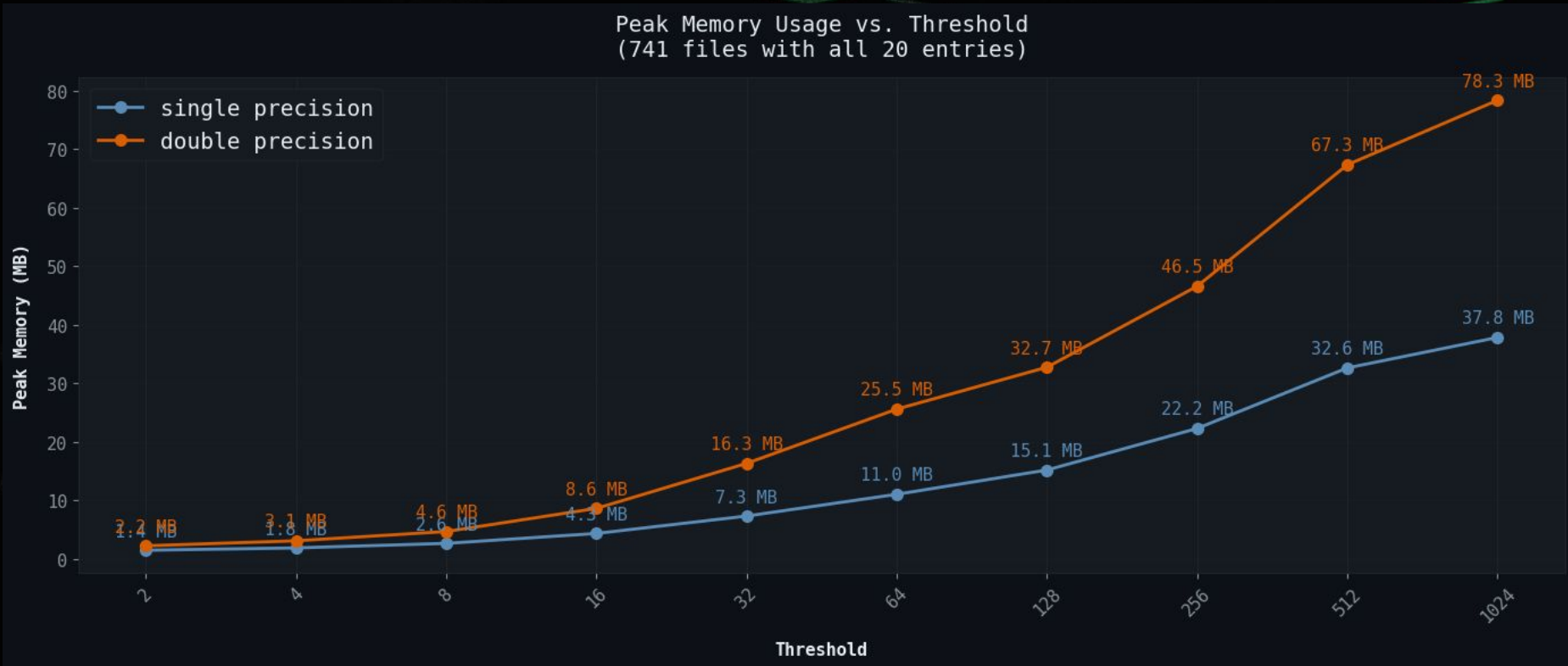
Runtime Distribution (~15,000 runs)



Precision & Threshold vs Runtime



Precision & Threshold vs Memory Use

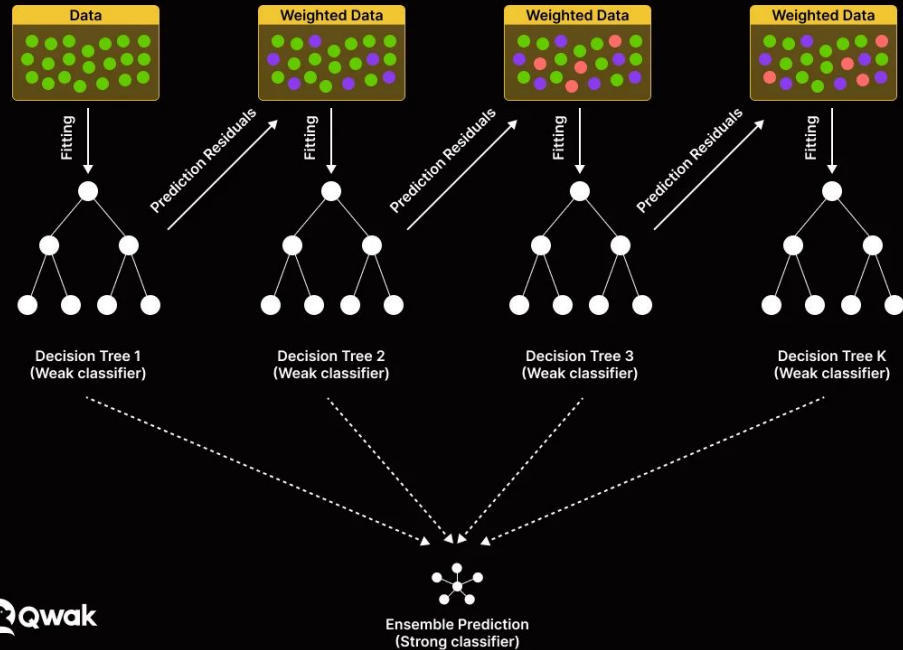


C

Model Results

Final Model: XGBoost Regressor

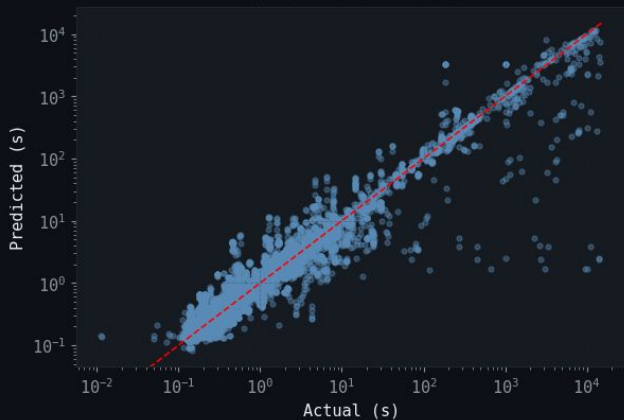
- Ensemble of decision trees
- Gradient boosting framework
- Regularization and error correction built-in
- Incredible fast and efficient



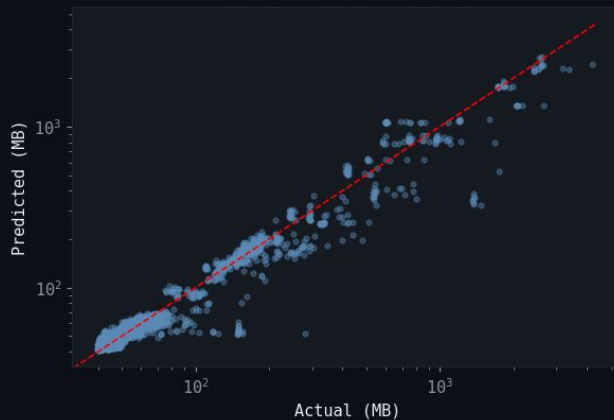
Final Model Results

True Out-of-Fold Cross-Validation: Predicted vs Actual

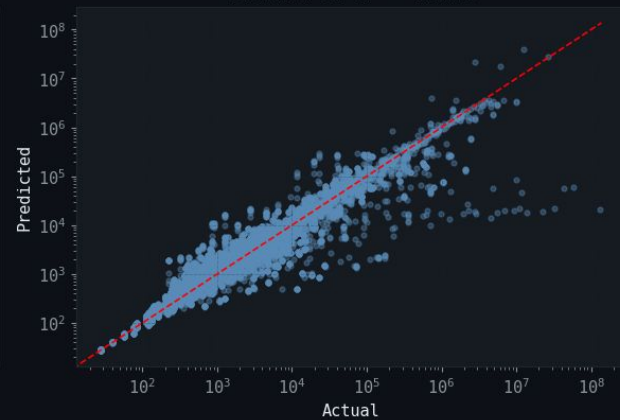
Runtime (s)
Overall CV $R^2 = 0.933$



Peak Program Memory (MB)
Overall CV $R^2 = 0.957$

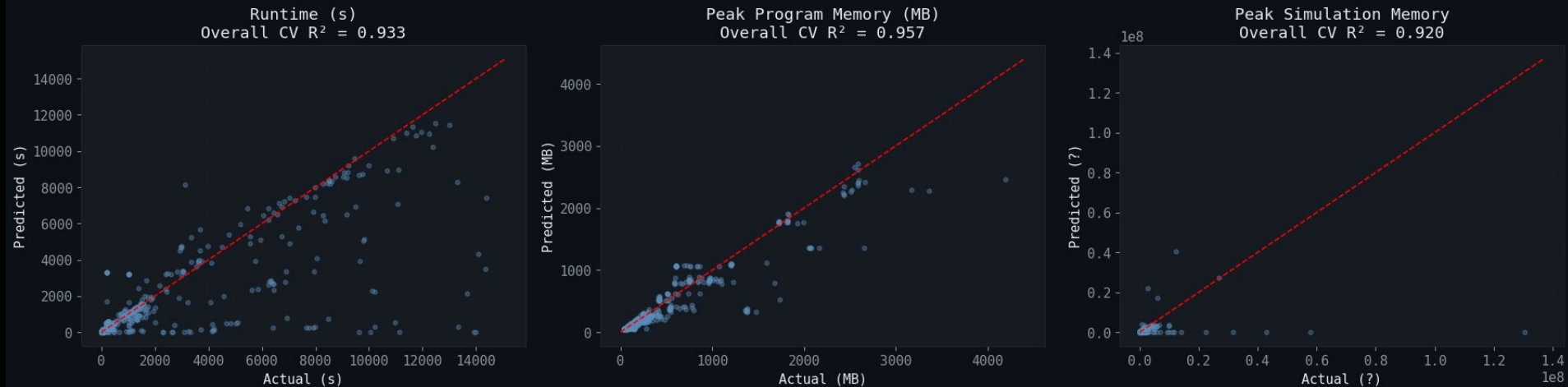


Peak Simulation Memory
Overall CV $R^2 = 0.920$



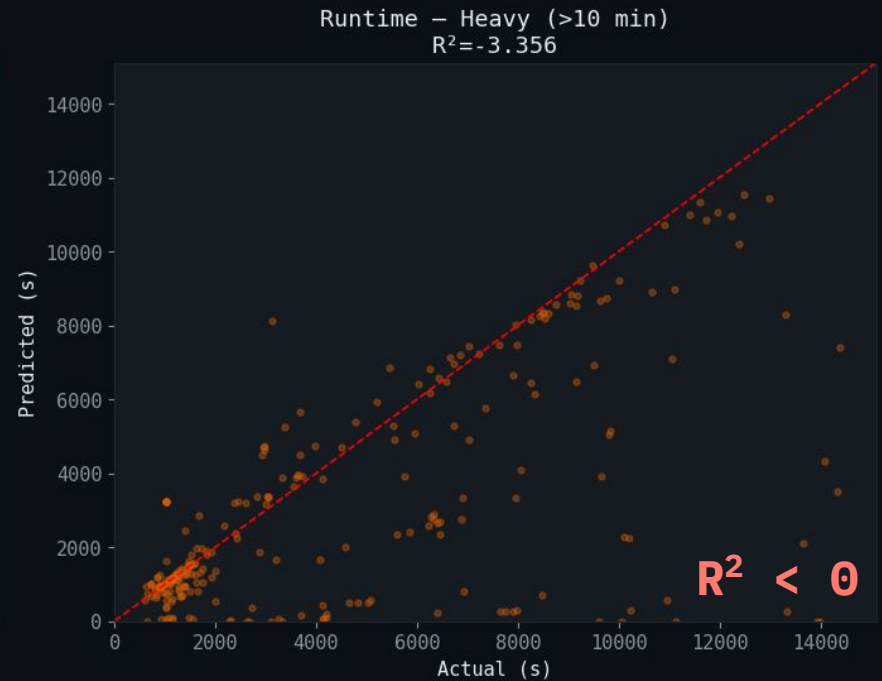
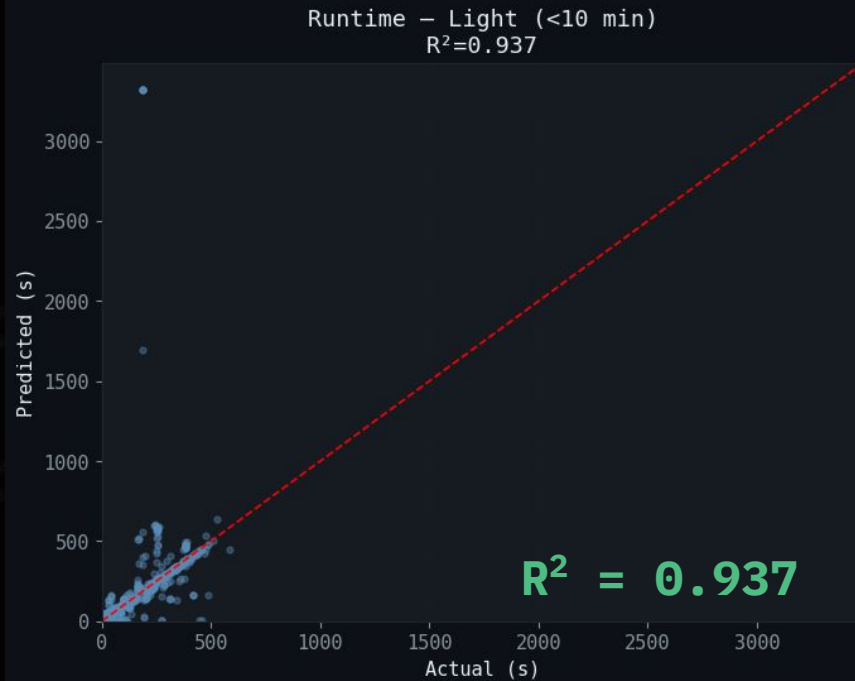
Final Model Results (True Scale)

True Out-of-Fold Cross-Validation: Predicted vs Actual



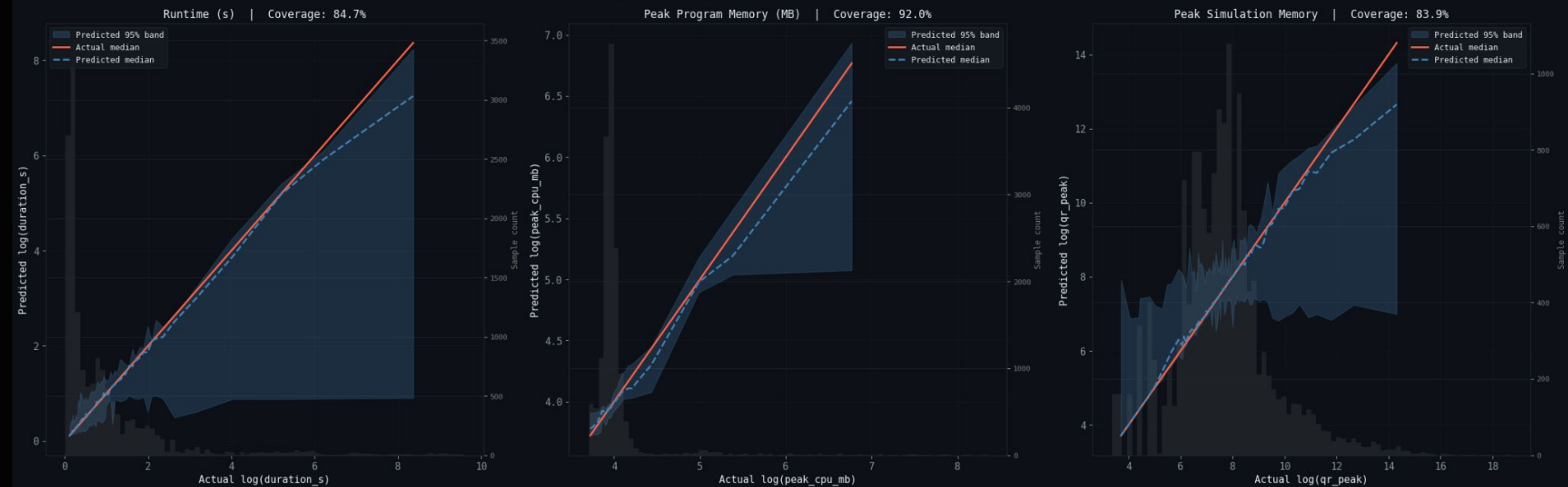
Where Accuracy Drops Off

Runtime OOF – Light vs Heavy Regime

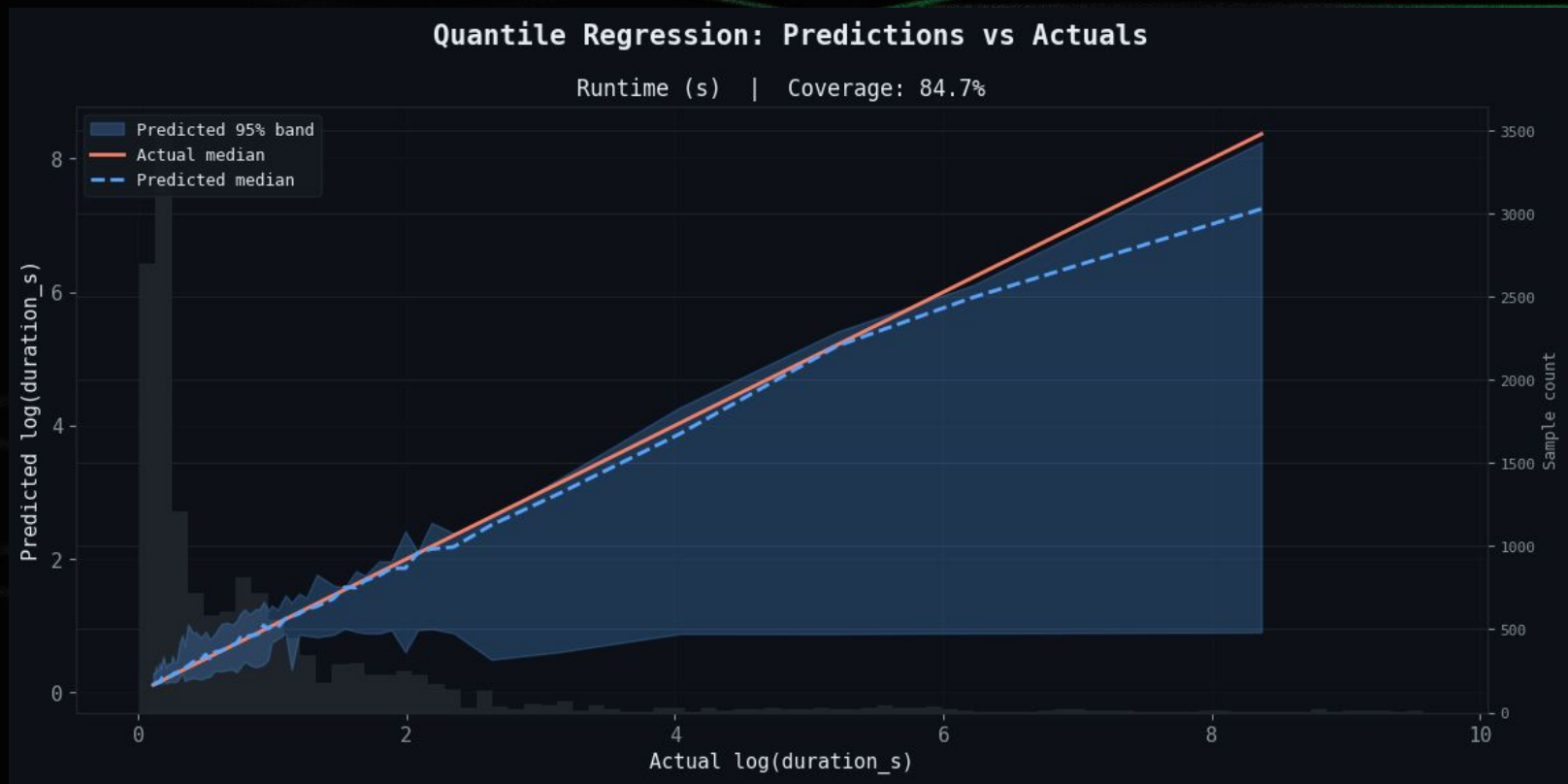


Quantile Model

Quantile Regression: Predictions vs Actuals



Quantile Model Runtime



$|011\rangle$

Demo

Quantum Rings
Quantum Circuit Predictor

Drag & drop .qasm or click

File: qftentangled_qubit50_IDM_Q127_EAGLE_best_gate_count_forward_swap_1004_gate_0667_fidelity-99.62753728658967.qasm

Precision: double Threshold: 256

Predict Performance

PREDICTION	GROUND TRUTH
RUNTIME 1.81 s Range: 1.81 ~ 10.12 s	RUNTIME (REAL) 6.78 s
PEAK MEMORY 58.94 MB Range: 46.43 ~ 73.63 MB	PEAK MEMORY (REAL) 58.59 MB
View Real Results	

Quantum Rings
Quantum Circuit Predictor

Drag & drop .qasm or click

File: grover_19.qasm

Precision: single Threshold: 512

Predict Performance

PREDICTION	GROUND TRUTH
RUNTIME 39.87 s Range: 1.99 ~ 53.92 s	RUNTIME (REAL) 381.71 s
PEAK MEMORY 741.48 MB Range: 115.52 ~ 986.88 MB	PEAK MEMORY (REAL) 833.34 MB
View Real Results	

Future Work

Where to go from here:

Expand
Dataset

(Focus on
Higher Runtime
Circuits)

Resolve GPU
Issues

(Include Amber
& Serin
Backends)

Collect
Data on
Variety of
Hardware

Increase
Shot Count
(1 → 1000)



Lessons Learned

Data Generation

- Basic understanding of circuit types
- Overcoming assumptions from classical compute

Quantum Computing

- Circuit format
- QASM features
- Industry

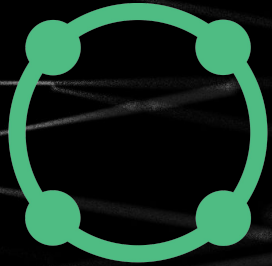
Wendian HPC

- Working in HPC clusters
- Benchmarking tasks
- Parallel computing

Machine Learning

- Fine-tuning models
- Identifying issues
- Exploring data





Quantum
Rings

Thank you!

Any questions?